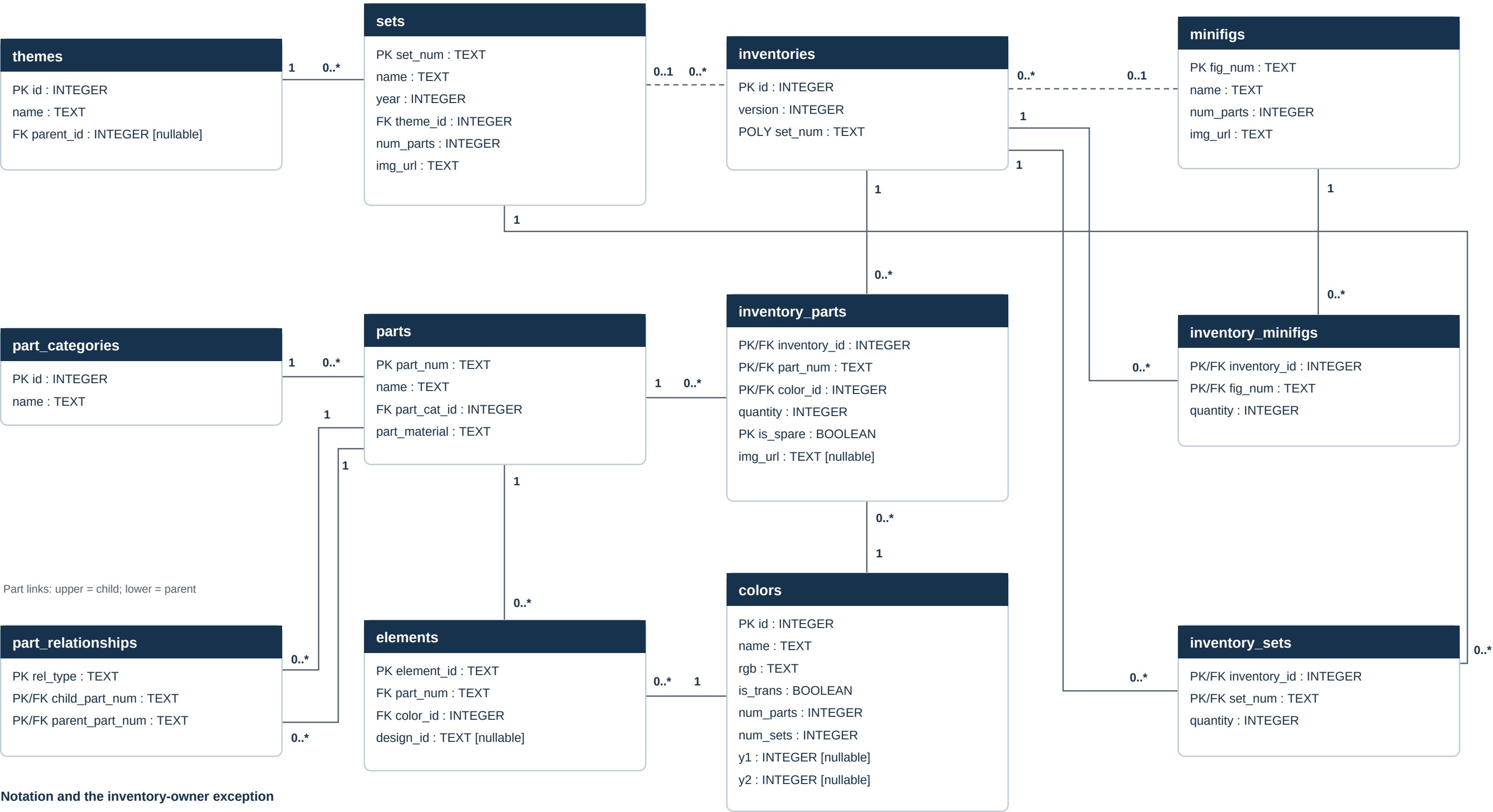


Task 1 | Actual Rebrickable CSV model

All 12 official CSVs inspected on 30 September 2026 | Logical keys inferred and validated from this downloaded snapshot.



Part links: upper = child; lower = parent

Notation and the inventory-owner exception

PK = inferred primary/candidate key; multiple PK attributes form one composite key. FK = reference validated against the snapshot. Numbers at line ends show UML multiplicities: 1 = exactly one; 0..1 = optional one; 0..* = zero or many. Theme hierarchy is self-referencing. Dashed owner links are alternatives (XOR): each inventory belongs to exactly one set OR one minifig. POLY set_num is not two ordinary SQL foreign keys. For inventories, (set_num, version) is an additional observed unique key. All other unmarked attributes are non-key CSV attributes. TEXT identifiers preserve letters, suffixes and leading zeroes. CSVs have no declared SQL types or constraints; displayed types are logical interpretations.

Task 1 | Findings and comparison

The downloaded CSV structure broadly matches the 12 entities in Figure 4, but it contains additional attributes and a polymorphic inventory-owner reference. The diagram on page 1 represents the complete source data, rather than the smaller application database required later in Task 3.

Method and provenance

Downloaded all 12 compressed CSV files from cdn.rebrickable.com/media/downloads/<file>.csv.gz on 30 September 2026. Inspected every header and every data row; tested the candidate keys for duplicate groups and checked nonblank references. The official download listing is rebrickable.com/downloads/. Snapshot hashes and results are provided in CSV_Inspection_Evidence.json.

Differences from assignment Figure 4

File / issue	Figure 4	Actual CSV / correction
parts.csv	part_num, name, part_cat_id	Also includes part_material.
sets.csv	set_num, name, year, theme_id, num_parts	Also includes img_url.
minifigs.csv	fig_num, name, num_parts	Also includes img_url.
inventory_parts.csv	inventory_id, part_num, color_id, quantity, is_spare	Also includes img_url; 6,931 rows have a blank URL.
elements.csv	element_id, part_num, color_id	Also includes design_id; 21,636 rows have a blank design_id.
colors.csv	id, name, rgb, is_trans	Also includes num_parts, num_sets, y1 and y2; two rows have blank y1/y2.
Inventory ownership	inventories.set_num is drawn as a reference to sets only.	30,368 inventory rows reference sets; 17,229 reference minifigs. Each row matches exactly one catalogue type.
Key and type semantics	Shows field types and links; does not establish CSV constraints.	Keys/types in this analysis are inferred and tested. CSV files themselves do not declare primary keys, foreign keys or SQL types.

No entities or columns shown in Figure 4 were missing from this snapshot. themes, part_categories, part_relationships, inventories, inventory_sets and inventory_minifigs retain the attributes shown in Figure 4.

Task 1 | Keys and validation

CSV file	Data rows	Validated candidate key
themes.csv	496	id
sets.csv	28,417	set_num
minifigs.csv	17,229	fig_num
inventories.csv	47,597	id
inventory_sets.csv	5,212	inventory_id, set_num
inventory_minifigs.csv	25,831	inventory_id, fig_num
inventory_parts.csv	1,558,562	inventory_id, part_num, color_id, is_spare
parts.csv	64,680	part_num
part_categories.csv	76	id
part_relationships.csv	37,438	rel_type, child_part_num, parent_part_num
elements.csv	114,315	element_id
colors.csv	275	id

Validation results

All 12 listed keys had zero duplicate groups. Every key component was nonblank. The additional inventory key (set_num, version) also had zero duplicate groups. All 14 ordinary reference checks had zero unmatched nonblank rows; inventory ownership also had zero unmatched values across the union of sets and minifigs.

These checks establish consistency in this snapshot, not permanent guarantees about future exports. Re-run them after refreshing the data. A single row quantity is a measure, not part of its key. is_spare belongs in the inventory_parts composite key so ordinary and spare copies of the same part and colour can remain separate.

Nullability and identifier handling

themes.parent_id is blank in 150 rows, representing root themes. The other blank fields are inventory_parts.img_url (6,931), elements.design_id (21,636), and colors.y1/y2 (two each). All remaining inspected fields were nonblank. Convert blank optional numeric fields to NULL on import; retain identifiers such as set_num, part_num, fig_num and element_id as text.

Task 1 | Relationship interpretation

Reference	Cardinality / meaning
themes.parent_id -> themes.id	A theme has 0..1 parent; a parent theme has 0..* child themes.
sets.theme_id -> themes.id	Each set has one theme; a theme has 0..* sets.
inventories.set_num -> sets.set_num OR minifigs.fig_num	Each inventory has exactly one catalogue owner (XOR). Each owner may have 0..* inventory versions in the logical model.
inventory_parts.inventory_id -> inventories.id	An inventory has 0..* part rows; each part row belongs to one inventory.
inventory_parts.part_num -> parts.part_num; color_id -> colors.id	Each part row references one part and one colour; each part/colour can appear in 0..* rows.
inventory_sets.inventory_id -> inventories.id; set_num -> sets.set_num	Each row references one containing inventory and one contained set; both parent entities can have 0..* such rows.
inventory_minifigs.inventory_id -> inventories.id; fig_num -> minifigs.fig_num	Each row references one containing inventory and one minifig; both parent entities can have 0..* such rows.
parts.part_cat_id -> part_categories.id	Each part has one category; a category has 0..* parts.
elements.part_num -> parts.part_num; color_id -> colors.id	Each element references one part and one colour; each part/colour has 0..* elements.
part_relationships.child_part_num / parent_part_num -> parts.part_num	Each relationship has one child part and one parent part; each part may appear in 0..* relationships in either role.

Interpretation cautions

inventories.set_num names the inventory owner. inventory_sets.set_num names a set contained within an inventory; these are different roles. Inventory contents and ownership must not be conflated.

Observed part relationship codes are A, B, M, P, R and T. Store these as categorical text rather than translating them into speculative rules. design_id is an exported identifier with blanks; no design table is included, so no foreign key is asserted for it.

For implementation, the polymorphic owner can be resolved using an owner catalogue or separate nullable owner references with an exclusive-or rule. That is a Task 3 design decision; the source diagram preserves the actual CSV field name.

Team review before submission

Review and significantly develop these AI-assisted drafts. Capture screenshots of the prompt and generated output in GitHub, as required by the assignment. Have each member commit their own work progressively. Confirm the dates, relationship notation and chosen data treatment together. No repository upload, Moodle submission or email is claimed by these documents.